

Point-Set Topology

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¹The relation $X \ll Y$ indicates that these notes were originally authored by Y and subsequently prepared for submission by X without significant modification and with appropriate permissions. While Y is credited with the genesis of these notes, X takes full responsibility for the current version. In this case, it indicates that these notes were originally authored by R. L. Moore

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To the Instructor

These notes are a rigorous one-semester Introduction to Topology Analysis with a prerequisite of Real Analysis I.

The notes are an augmented version of the notes I took when I studied under Michel Smith at Auburn University in the 1980s whilst his notes originated from his studies of Topology under W. S. Mahavier at Emory University who studied under R. L. Moore at the University of Texas. Whilst I was a faculty member at Morehouse College (1988 - 2005) I used these notes and I have used them at Kutztown University (2005 - present). I have not changed them (if any very little) from the notes I took under Michel Smith. I believe they are, perhaps, the mostly unaltered notes on Point-Set Theory that were originated by Moore.

The course is a 400-level (senior-graduate student) course at Kutztown University. The students are not required to take this course the major nor for the M.S. Mathematics Secondary Education major. Typically it is offered once every three years at Kutztown with an enrolment of approximately 10 (some years as few as 5 and as much as 16). We hold the class in the Inquiry Based Learning (IBL) classroom (moveable tables and chalkboards on every wall).

I teach the course in a strict IBL style where the student is graded on presentations (clarity of the work; ease with the concepts; ability to answer questions and defend their thoughts; but, also a willingness to hear feedback and correction when necessary) and participation (the questions they ask of the presenter or instructor; the comments they make; and, the willingness to present). The rules of the class are simple – no collaboration or discussion outside of class before presentation; no use of outside resources; and humility & humour are suggested.

I accentuate an axiomatic approach requiring students to explain in detail why the student deduces a statement. I encourage the students to think less literally (just regurgitation) and more geometrically.

I do not find the need for tests most of all because almost all the students who take the class have had me in Calculus, Set Theory, Real Analysis, or Probability & Statistics where I use a more relaxed version of IBL. So, they are accustomed to the rules about my IBL classes and seek out the course rather than being required to take it.

I ended the notes with a taste of product spaces the continuum hypothesis and indecomposable continua to give the student a ‘preview’ of material that they

would encounter if they were to continue studying Point-Set Theory. At each of the institutions where I have taught we do not have a second course in Topology or a graduate programme; but, I have directed advanced students in Point-Set Theory intending to go on to graduate school in independent study.

To the Student

These notes are a rigorous one-semester Introduction to Topology with a prerequisite of Real Analysis I.

The axioms, definitions, lemmas, theorems, corollaries, and claims proceed in a manner such that it is not necessary to use say theorem X to prove theorem X + 5; but, it may be of use.

These notes contain more than sufficient material for a one-semester undergraduate topology course or a one-semester first year graduate course. The notes cover an essential introduction to Point-Set Theory. The notes offer the student the opportunity to investigate limit points, closed or open sets, compactness, countability, sequences, continuity, connectedness, separability, local compactness, and total connectedness.

Rules governing this class:

1. you are **prohibited** from using textbooks during the learning process;
2. we will have but the briefest of lectures in class;
3. **no collaboration** or conferring between classmates;
4. you are **not** to discuss your work with students not in this class;
5. you are **not** to discuss your work with other Professors;
6. advice or input is from the instructor or you alone - that is it; and,
7. after a claim, lemma, theorem, corollary, exercise (whatever) has been presented and is deemed correct you may discuss the solution, proof, counter-example with anyone in this class but do not do so with students not in this class or with other Professors (except, perhaps, that it was discussed in class) but try not to spoil it for one who has not taken this class and try not to let someone else spoil it for you by providing commentary beyond the scope of the material as we are learning it - there are some wonderful people in the department - very enthusiastic, chatty people - who can inadvertently ruin a class for a person by revealing 'too much';

The point of this exercise is to transform you from recipients of information to creators of knowledge. It is in essence a Socratic method that encourages students to solve problems using your own skills of critical analysis and creativity.

Chapter 1

Foundational Topology

We will begin with some reiterations of previous definitions (now renumbered - they are so important) and some foundations work.

Assumption A 1.1. The term “point” and “region” are undefined and every region is a point-set.¹

Notation A 1.2. S denotes the set of all points and S is called the **space**.

Definition A 1.3. Let S be a space. If M and W are point-sets such that they have no common elements then we say M and W are **mutually exclusive** or **disjoint**.

Notation A 1.4. When M and W are mutually exclusive point-sets we use the notation $M \cap W = \emptyset$.²

Definition A 1.5. Let S be a space. If M is a point set and p is a point then the statement p is a **limit point** of M means that every region containing p also contains a point of M distinct from p .

Definition A 1.6. Let S be a space. If M is a point set then $\overline{M}$ denotes the point set such that p belongs to it iff p is a point of M or p is a limit point of M . The point set $\overline{M}$ is called **the closure of M** .

Definition A 1.7. Let S be a space. The collection $\mathcal{G}$ of point sets is said to **cover** the point set M iff each point in M belongs to a point set in $\mathcal{G}$.

¹The terminology point-set shall mean that any point-set is composed of at least one point.

²For a set theorist who accepts the axiom of null; this is a set. For the set theorist who rejects the axiom of null it is but a notational norm and stands for $M \cap W$ does not exist.

Axiom A. Let S be a space. There exists a sequence $G_1, G_2, G_3, \dots$ such that:

1. for each positive integer n , G_n is a collection each element of which is a region;
2. for each positive integer n , G_n covers S and G_{n+1} is a sub-collection of G_n ;
3. if R is a region, a is a point of R , b is a point of R , then there exists a positive integer m such that if G is a region in G_m containing a then $\overline{G}$ is a subset of R , and if b is distinct from a , then $\overline{G}$ does not contain b .

Definition A 1.8. M is a point set and p is a point. p is a **boundary point** of M iff every region containing p contains a point in M and a point not in M .

Definition A 1.9. Let M be a point set. The set of boundary points of M is called as the **boundary** of M and is denoted as $Bd(M)$.

Theorem A 1.10. No point of a region is a boundary point of the region.

Definition A 1.11. If M is a point set then the statement M is **closed** means that there is not a limit point of M that does not belong to M .

Theorem A 1.12. If M is a point set then $\overline{M}$ is closed.

Axiom T. Every point lies in some region and if the point p lies in each of the regions R_1 and R_2 then there is a region lying in the common part of regions R_1 and R_2 that contains p .

Axiom H. If p and q are two distinct points, then there are mutually exclusive regions: one containing p and the other containing q .

Exercise A 1.13. Axiom A implies Axiom T and Axiom H.

Exercise A 1.14. Give one example of a space S that satisfies Axiom T and Axiom H (please define “point” and “region” for your example).

Theorem T 1.15. If X and Y are point sets then the following holds true on the basis of Axiom T:

1. $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$
2. $\overline{\overline{X}} = \overline{X}$

Question T 1.16. Does the converse of theorem T 1.15 hold?

Definition A 1.17. The space S is said to be a T_0 - space provided that for each pair of distinct points there is a region that contains one of the points, but not the other.

Definition A 1.18. The space S is said to be a T_1 - space iff S satisfies Axiom T together with the condition that a singleton point set is closed.

Comment A 1.19. A singleton point set $\{p\}$ is closed means $\{p\} = \overline{\{p\}}$.

Definition A 1.20. A point set O is said to be **open** iff for each point p lying in O there exists a region containing p that is a subset of O .

Definition A 1.21. The space S is said to be a T_2 - space means that S satisfies Axiom H. (Note: it is also called a **Hausdorff space**).

Definition A 1.22. Suppose the space S is a T_1 - space, then S is said to be **regular** iff it is true that if p is a point of S and if F is a closed point set not containing p then there exists disjoint open sets U and V such that $p \in U$ and $F \subseteq V$.

Definition A 1.23. The space S is said to be a T_3 - space means that S is a regular T_1 - space .

Exercise A 1.24. Suppose the space S satisfies Axiom A. For which integers $n = 0, 1, 2, 3$ is S a T_n - space?

Exercise A 1.25. For what integers m and n is the statement “if S is a T_m - space, then S is a T_n - space” true?

Axiom C. Suppose $\mathcal{G}$ is a collection of point sets, then there exists a function f from $\mathcal{G}$ into S so that for each $R \in \mathcal{G}$ it is the case that $f(R) \in R$.

Example C 1.26. $\mathcal{G} = \{\{1, 2, 3, 4\}, \{3, 4\}, \{1\}, \{3, 7\}\}$ where $A = \{1, 2, 3, 4\}, B = \{3, 4\}, C = \{1\}, D = \{3, 7\}$. $f(A) = 2, f(B) = 4, f(C) = 1$, and $f(D) = 7$

Notation C 1.27. Suppose that M is a point set and there exists a meaning for the word precedes “ $\prec$ ” such that each of x and $y \in M$, then we symbolise ‘ x precedes or is the same as y ’ as $x \preceq y$ and if ‘ x precedes and is not is the same as y ’ as $x \prec y$ or with any other symbol that communicates the concept and is well-defined.

Definition A 1.28. Suppose that M is a point set and there exists a meaning for the word precedes “ $\prec$ ” such that each of x, y , and $z \in M$, then:

1. if $x \neq y$ then $x \prec y$ or $y \prec x$;
2. if $x \prec y$ and $y \prec z$, then $x \prec z$;
3. if $x \prec y$ then $y \not\prec x$; and,
4. if $K \subseteq M$, then there exists an element of K which is preceded by no other element of K .

then M is said to be **well ordered** with respect to the meaning of the word “precedes.”

Definition A 1.29. The collection $\mathcal{G}$ of point sets is said to be **monotonic** if it is true that if $X \in \mathcal{G}$ and $Y \in \mathcal{G}$, then $X \subseteq Y$ or $Y \subseteq X$.

Definition A 1.30. The point set M is said to be **perfectly compact** iff it is true that if $\mathcal{G}$ is a monotonic collection of subsets of M , then there exists a point p so that p is either belongs to every set in the collection $\mathcal{G}$ or p is a limit point of every set in the collection $\mathcal{G}$.

Theorem A 1.31. If the point set M is closed and perfectly compact and $\mathcal{G}$ is a collection of regions which is well ordered with respect to some meaning of the word precedes, Q , and collection $\mathcal{G}$ covers M then there exists a region R in the collection $\mathcal{G}$ such that M is covered by the collection whose elements are R together with the elements of $\mathcal{G}$ that precede R with respect to Q .

Definition A 1.32. Suppose M is a point set such that if $\mathcal{G}$ is a collection of regions that covers M , then there exists a finite sub-collection, $\mathcal{H}$ of $\mathcal{G}$ that covers M . Then, M is said to be **compact**.

Question A 1.33. Let S be a space. Is it the case that if M is a compact point set then M is closed assuming:

1. Axiom A (e.g.: the space S that M is in satisfies Axiom A);
2. Axiom T (e.g.: the space S that M is in satisfies Axiom T);
3. Axiom H (e.g.: the space S that M is in satisfies Axiom H);
4. Axiom T and Axiom H (e.g.: the space S that M is in satisfies both Axiom T & H);
5. the space S is a T_1 - space.

Question A 1.34. Does there exist a space S and a point-set that is a proper subset of S , $M \subset S$, such that M is both open and closed.

1. Axiom A (e.g.: the space S that M is in satisfies Axiom A);

2. Axiom T (e.g.: the space S that M is in satisfies Axiom T);
3. Axiom H (e.g.: the space S that M is in satisfies Axiom H);
4. Axiom T and Axiom H (e.g.: the space S that M is in satisfies both Axiom T & H);
5. the space S is a T_1 - space.

Comment A 1.35. A space S which is a regular Hausdorff space and which satisfies Axiom A is called a **Moore space**.

Question A 1.36. Does there exist a space S and a point-set that is a proper subset of S , $M \subset S$, such that if M is open, then $S \setminus M$ is closed assuming:

1. Axiom A (e.g.: the space S that M is in satisfies Axiom A);
2. Axiom T (e.g.: the space S that M is in satisfies Axiom T);
3. Axiom H (e.g.: the space S that M is in satisfies Axiom H);
4. Axiom T and Axiom H (e.g.: the space S that M is in satisfies both Axiom T & H);
5. the space S is a T_1 - space.

Definition A 1.37. If S is a space and $\mathcal{T}(S)$ is the collection to which O belongs iff O is open, then $\mathcal{T}$ is referred to as **the topology** of S .

Question A 1.38. Suppose S is a set (the space we define), does there exist a meaning for the word “region” with respect to which S together with the word region that satisfies Axiom A?

Theorem C 1.39. Let S be a space. If M is a point set, then there exists a meaning for the word precedes with respect to which M is well ordered.

From this point on, when a theorem is titled “Theorem A.something,” you are to assume both Axiom A and Axiom C (theorem C 1.39).

Question C 1.40. If you have theorem C 1.39 as an Axiom (say Axiom C_1), then does Axiom C follow from Axiom C_1 as a theorem?

Lemma A 1.41. Let S be a space. Suppose $\mathcal{G}$ is a collection of regions covering M and $\mathcal{G}$ is well ordered, then $\forall K \in \mathcal{G}$ it is the case that $M - K$ is perfectly compact and closed.

Theorem A 1.42. Let S be a space. If M is a closed and perfectly compact point set, then M is compact.

Theorem A 1.43. Let S be a space. If M is compact, then M is perfectly compact.

Exercise A 1.44. Show that there exists a meaning, Q , for the word region so that the space $\mathbb{R}$ satisfies Axiom A, so that a point x is a limit point of the set M with respect to the meaning Q iff x is a limit point by the “standard” definition of limit point.

Exercise A 1.45. Show that there exists a meaning, Q , for the word open set so that the space $\mathbb{R}$ satisfies Axiom A, so that a point x is a limit point of the set M with respect to the meaning Q iff x is a limit point by the “standard” definition of limit point.

Comment A 1.46. The previous exercises are intended to note that $\mathcal{T}(S)$ with respect to Q is $\mathcal{T}(S)$ with respect to a “standard” definition.

Exercise A 1.47. Let S be a space. If M is a point set, then the boundary of M is closed.

Exercise A 1.48. Let S be a space. If M is a compact point set and p is a point not in M , then there exists disjoint open sets U and V such that $p \in U$ and $M \subseteq V$.

Exercise A 1.49. Let S be a space. If M is a compact point set and K is a closed non-empty subset of M , then K is compact.

Question A 1.50. For what positive integers m and n is the statement a “ T_m - space is a T_n - space” provided we assume S is a compact T_m - space?

Question A 1.51. Let S be a Moore space. If M is closed, is M compact?

Question H 1.52. Let S be a Hausdorff space. If M is closed, is M compact?

Comment A 1.53. Let U be a well-defined universe in set theory and A a set. Recall the definitions of **null**, **finite**, **infinite**, **denumerable**, **countable**, and **uncountable** sets.

Theorem A 1.54. Let S be a Moore space. Suppose M is an uncountable point set such that every uncountable subset of M has a limit point. Suppose that the word “precedes” (“ $\preceq$ ”) has such a meaning that:

1. if x and y are two distinct points of M then either $x \preceq y$ or $y \preceq x$;
2. if x and y are two distinct points of M and $x \preceq y$ then $y \not\preceq x$;

Suppose furthermore y is said to follow x iff $x \preceq y$, then some point p is a limit point of the set of points that precede p and of the set of points that follow p .

Theorem A 1.55. Let S be a Moore space. If M is compact, then every infinite subset of M has a limit point.

Theorem A 1.56. Let S be a Moore space. If p is a point, then there exist an infinite sequence $R_1, R_2, R_3, \dots$ such that:

1. for each $n \in \mathbb{N}$, R_n is a region with $\overline{R_{n+1}} \subseteq R_n$;
2. p is the only point that belongs to every region of the sequence; and,
3. if R is a region containing p , then there exists a positive integer k such that $\overline{R_k} \subseteq R$.

Definition A 1.57. If S is a topological space with some $\mathcal{T}(S)$ and S satisfies the conclusion of Theorem A 1.56 for every point $p \in S$, S is said to be **first countable**.

Question A 1.58. Does there exist a T_2 - space that is not first countable?

Theorem A 1.59. Let S be a Moore space. If H is a compact point set and K is a closed point set such that they are disjoint, then there exists disjoint open sets U and V such that $H \subseteq U$ and $K \subseteq V$.

Definition M 1.60. Suppose S is a space, then the function $d : S \times S \longrightarrow [0, \infty)$ is called a **metric** on S iff:

1. if $x \in S$ and $y \in S$, then $d((x, y)) \geq 0$ and $d((x, y)) > 0$ when $x \neq y$;
2. if $x \in S$ and $y \in S$, then $d((x, y)) = d((y, x))$; and
3. if $x \in S$, $y \in S$, and $z \in S$, then $d((x, z)) \leq d((x, y)) + d((y, z))$.

Definition M 1.61. The topological space S is said to be a **metric space** iff there exists a metric, d , on S such that the collection $\mathcal{G}$ to which the set O belongs iff for each point $p \in O$ there exists an $\varepsilon > 0$ so that $\{x | d((p, x)) < \varepsilon\}$, is a subset of O , is the collection $\mathcal{T}(S)$.

Note A 1.62. Definition M 1.61 implies that a metric induces a topology.

Theorem M 1.63. If S is a metric space then S satisfies Axiom A (e.g.: S is an Axiom A space for some meaning of the word region).

Theorem A 1.64. Let S be a Moore space. If p is a limit point of the point set M , then every region containing p contains infinitely many elements of M .

Definition A 1.65. Let S be a Moore space. If $\langle p_1, p_2, p_3, \dots, p_i, \dots \rangle$ where $i \in \mathbb{N}$ is a sequence each element of which is a point, then p is **the sequential limit of the sequence** means that for each region R containing p there exists a positive integer n so that if $k > n$ and k is an integer, then p_k lies in R .

Exercise A 1.66. Let S be a Moore space. No sequence has two sequential limits.

Theorem A 1.67. Let S be a Moore space. If p is a limit point of the point set M , then there exists an infinite sequence of distinct points of M (all distinct from each other and from p) such p is the sequential limit of the sequence.

Theorem A 1.68. Let S be a Moore space. If for each positive integer n p_n is a point; M is the set of all points $\{p_n | n \in \mathbb{N}\}$ and M has the property that every infinite subset of M has a limit point, then there exists an increasing sequence of positive integers $\langle n_1, n_2, n_3, \dots \rangle$ such that $\langle p_{n_1}, p_{n_2}, p_{n_3}, \dots, p_{n_i}, \dots \rangle$ has a sequential limit point.

Exercise A 1.69. Let S be a Moore space. Suppose that M is well ordered, then no increasing sequence positive integers $\langle n_1, n_2, n_3, \dots \rangle$ such that $\langle p_{n_1}, p_{n_2}, p_{n_3}, \dots, p_{n_i}, \dots \rangle$ has the property that p_{n+1} precedes p_n for all $n \in \mathbb{N}$.

Exercise A 1.70. Suppose that S is well ordered and there is an element of S which is preceded by every other element of S . Suppose further that region has the following meaning so that R is a region iff:

1. there exists a point a so that $R = \{x | x < a\}$;
2. there exist points a and b so that $R = \{x | a < x < b\}$; or,
3. there exists a point b so that $R = \{x | x > b\}$.

then S is compact.

Question A 1.71. (related to exercise 1.70) For what integer n is S a T_n - space from the previous exercise?

Claim A 1.72. If M is well ordered and every subset of M has a greatest element, then M is finite.

Definition A 1.73. The topological space S is said to be **normal** iff for each of two disjoint closed sets H and K there exist disjoint open sets U and V such that $H \subseteq U$ and $K \subseteq V$.

Question A 1.74. Does there exist an Axiom A space that is not normal?

Definition A 1.75. Suppose each of S_1 and S_2 is a topological space and f is a function from S_1 into S_2 , f is said to be **continuous** iff for each $x_1 \in S_1$ and for each open set $R \subseteq S_2$ containing $f(x_1)$ there exists an open set U ($U \subseteq S_1$ obviously) containing x_1 so that $f(p) \in R \forall p \in U$.

Exercise A 1.76. Suppose each of S_1 and S_2 is a topological space and f is a well defined function, $f : S_1 \longrightarrow S_2$, from S_1 into S_2 , then f is continuous iff for each open set R , a subset of S_2 , $f^{-1}(R)$ is an open set ($f^{-1}(R) \subseteq S_1$, logically) where $f^{-1}(R) = \{x | f(x) \in R\}$.

Definition A 1.77. Let $S = \mathbb{R}$. A **segment** is a point-set such that either:

1. there exists two distinct points a and b where $a < b$ such that the segment is the set $\{x | a < x < b\}$;
2. there exists a point c such that the segment is the set $\{x | x < c\}$; or,
3. there exists a point d such that the segment is the set $\{x | x > d\}$.

Exercise A 1.78. Let $S = \mathbb{R}$. Show for $[0, 1]$ the definition of open set implies that segments intersecting $[0, 1]$ are open sets.

Theorem A 1.79. The topological space S is normal iff for each of pair of disjoint closed sets H and K there exists a continuous function f from S into $[0, 1]$ so that $f(h) = 0 \forall h \in H$, $f(k) = 1 \forall k \in K$ ($f[H] = \{0\}$, $f[K] = \{1\}$).

Definition A 1.80. If M is a well ordered set and N is a well ordered set, where we have $\langle M, \preceq \rangle$ and $\langle N, \prec \rangle$, then M and N are **order isomorphic** if there exists a bijection $f : M \rightarrow N$ such that if $x_1, x_2 \in M$ and $x_1 \overset{M}{\preceq} x_2$ then it is the case that $f(x_1) \overset{N}{\prec} f(x_2)$ (where, of course, $f(x_1), f(x_2) \in N$).

Definition A 1.81. If M and N are order isomorphic, then the bijection f establishing this is called an **order isomorphism**. We denote such oft as $M \xrightarrow{f} N$

Definition A 1.82. If M is a well ordered set under a well-defined order, $<$, I is said to be an **initial segment of M** means that there is an element $x \in M$ so that $I = \{z | z < x\}$.

Exercise A 1.83. Suppose M is a well ordered set, show there exists a well ordered set $\hat{M}$ so that $\hat{M}$ is not order isomorphic M nor to any initial segment M .

Exercise A 1.84. Suppose M is a well ordered set and N is a well ordered set, then either M is order isomorphic to N , or there exists an initial segment of one which is order isomorphic to the other.

Exercise A 1.85. Suppose N is a well ordered set, then N is not order isomorphic to any initial segment of N (under the same meaning of the word precedes).

Definition A 1.86. If H is a set and K is a set, then H is said to be **equally numerous** to K means there exists a one-to-one function from H onto K (a bijection between H and K).

Comment A 1.87. Two sets that are equally numerous are called **equinumerous**; **equipotent**; or, **equipollent**.

Definition A 1.88. If H is a set and K is a set, then H is said to be **less numerous** than K means:

1. H is not equally numerous to K ;
2. H is equally numerous to a subset of K .

Theorem C 1.89. If H , K , and L are sets and H is less numerous than K , whereas K is less numerous than L , then H is less numerous than L .

Lemma A 1.90. Suppose S is a topological space. Under the hypothesis of theorem A1.54, If L is an uncountable subset of M , then L contains one of its limit points (a limit point of L).

Theorem A 1.91. Suppose S is a topological space. Suppose H is a point set and K is a point set so that they are both closed and disjoint and $\mathcal{G}$ is a monotonic collection of closed and perfectly compact point sets so that each element of $\mathcal{G}$ intersects both H and K , but is not the union of two closed disjoint point sets one intersecting H and the other intersecting K . Then, then common part, M , of the elements of $\mathcal{G}$ ($M = \bigcap \mathcal{G}$) has the same property (that is, intersecting both H and K but is not the union of two closed disjoint point sets one intersecting H and the other intersecting K).

Definition A 1.92. Suppose S is a topological space. If H is a point set and K is a point set, then H and K are said to be **mutually separated** iff no one of them contains a point or limit point of the other.

Definition A 1.93. Suppose S is a topological space. If M is a point set, then M is said to be **connected** iff M is not the union of two mutually separated points sets.

Definition A 1.94. Suppose S is a topological space. If M is a point set, then M is said to be a **continuum** if M is both compact and connected.

Definition A 1.95. Suppose S is a topological space. If M is a point set, then M is said to be **degenerate** if M has only one element.

Comment A 1.96. Suppose U is a well-defined universe. Recall from Set Theory, we say the set A is **trivial** if $A = \emptyset$ or $A = U$.

Exercise A 1.97. Suppose S is a topological space. If M is a non-degenerate connected point set, then every point of M is a limit point of M .

Theorem A 1.98. Suppose S is a topological space. If H is a point set and K is a point set whilst H and K are mutually separated, whereas M is a connected subset of $H \cup K$, then either M is a subset of H and M not a subset of K or M is a subset of K and M is not a subset of H ; but, not both.

Theorem A 1.99. Suppose S is a topological space. If M is a connected set and L is a subset of the set of limit points of M , then $M \cup L$ is connected.

Theorem A 1.100. Suppose S is a topological space. If $\mathcal{G}$ is a collection of connected point set and one of the members of the collection contains a point or limit point of all the members of the collection, then the union of all the members of $\mathcal{G}$ is connected.

Theorem A 1.101. Suppose S is a topological space. If $\mathcal{G}$ is a monotonic collection of continua, then the common part of the members of the collection is a continuum.

Question A 1.102. Suppose S is a topological space. Is it true that if $\mathcal{G}$ is a monotonic collection of connected point sets with a common part, then the common part of the members of the collection is connected?

Theorem C 1.103. Suppose S is a topological space. If $\mathcal{G}$ is a collection of sets such that if $\mathcal{H}$ is a monotonic sub-collection of $\mathcal{G}$, and there is a set in $\mathcal{G}$ which is a subset of every set in $\mathcal{H}$, then there is a set in $\mathcal{G}$ which contains no other set in $\mathcal{G}$.

Definition A 1.104. Suppose S is a topological space. If H and K are point sets, then the continuum M is said to be **irreducible** from H to K means that no proper sub-continuum M intersects H and K .

Theorem A 1.105. Suppose S is a topological space. If N is a compact point set and N intersects each of two disjoint point sets H and K , and if N is not the union of two disjoint closed point sets one intersecting H and the other intersecting K , then N contains a continuum irreducible from K to H .

Exercise A 1.106. Suppose S is a topological space. Find a continuum M that is irreducible.

Exercise A 1.107. If K is a connected set; $\hat{S}$ is a point set; and, $f : K \longrightarrow \hat{S}$ is continuous, then $f[K]$ is a connected set.

Exercise A 1.108. If K is a continuum; $\hat{S}$ is a set of points; and, $f : K \longrightarrow \hat{S}$ is continuous, then $f[K]$ is a continuum.

Question A 1.109. Does there exist a non-degenerate connected set which does not contain the one-to-one continuous image of $[0, 1]$?

Exercise A 1.110. Is $(0, 1)$ the one-to-one continuous image of $[0, 1]$?

Definition A 1.111. The point set M is said to be **locally compact** iff for each point $x \in M$, there exists an open set U containing x such that $\bar{U} \cap M$ is compact.

Definition A 1.112. Suppose S is a topological space. The point-sets H and K are said to be **topologically equivalent** or **homeomorphic** iff there exists a function that is continuous and onto from H to K that is invertible with a continuous inverse function from K onto H .

Question A 1.113. Suppose S is a topological space. If the point-sets H and K are homeomorphic then must there exist a function between them that is bijective (e.g.: must they be equinumerous)?

Theorem A 1.114. No locally compact point set M is the union of countably many closed point sets so that if G is any one of the closed point sets, then every point of G is a limit point of $M \setminus G$.

Lemma C 1.115. If M is a set then there exists a well ordering for M such that no initial segment of M is equally numerous with M .

Exercise A 1.116. Is the following statement true or false?

“If $\mathcal{G}$ is a monotonic collection of compact sets and P is the common part of the elements of $\mathcal{G}$ whilst $K \in \mathcal{G}$, then there is a set $C \in \mathcal{G}$ so that C does not intersect $K \setminus P$ (e.g.: $C \cap (K - P) = \emptyset$).”

Exercise A 1.117. Refer back to exercise A1.70. Determine if a point set M in such a space S has any of the following properties:

1. first countable;
2. normal;
3. satisfies Axiom A;
4. locally compact;
5. connected;
6. perfectly compact.

Comment A 1.118. Let $\varepsilon \in \mathbb{R}$ be given such that $\varepsilon > 0$. Please note:

1. that the standard topology for $\mathbb{R}$ means that regions are segments. Region $_{p,\varepsilon} = \{x | d(x, p) < \varepsilon\}$ such that d is the standard distance between points on the line (e.g.: let x and p be points $d(x, p) = |x - p| = \sqrt{(x - p)^2}$);
2. that the standard topology for $[0, 1]$ means that regions are common parts of segments of $\mathbb{R}$ with $[0, 1]$;
3. that the standard topology for $\mathbb{R} \times \mathbb{R} (\mathbb{R}^2)$ means that regions are interiors of circular discs. Region $_{p,\varepsilon} = \{x | d(x, p) < \varepsilon\}$ such that d is the standard distance between points in the plane (e.g.: x is (x_1, x_2) and p is (p_1, p_2) so $d(x, p) = \sqrt{(x_1 - p_1)^2 + (x_2 - p_2)^2}$);
4. that the standard topology for $\mathbb{R}^3$ means that regions are interiors of spherical space. Region $_{p,\varepsilon} = \{x | d(x, p) < \varepsilon\}$ such that d is the standard distance between points in 3-space (e.g.: x is (x_1, x_2, x_3) and p is (p_1, p_2, p_3) so $d(x, p) = \sqrt{(x_1 - p_1)^2 + (x_2 - p_2)^2 + (x_3 - p_3)^2}$);
5. etc.

Definition A 1.119. If $\mathcal{G}$ is a collection, then $\mathcal{G}^* = \bigcup \{A | A \in \mathcal{G}\}$.

Exercise C 1.120. Show that Theorem C1.39 follows from Theorem C1.103 and that Theorem C1.103 follows from Theorem C1.39.

Definition A 1.121. If M is a point set, then C is a **component of M** means that C is connected and is a proper subset of no connected subset of M .

Definition A 1.122. The point set M is said to be **totally disconnected** iff no non-degenerate subset of M is connected.

Definition A 1.123. If M is a point set, then the **derived set of M** means the set of limit points of M and is written as $Dr(M)$.

Theorem A 1.124. If $\mathcal{H}$ is a countable collection of compact sets and the collection has more than one element such that no non-degenerate component of an element of $\mathcal{H}$ intersects another element of $\mathcal{H}$, then $\mathcal{H}^*$ is not a locally compact, closed, and connected set.

Exercise A 1.125. Every point belongs to a component so long as the point belongs to a point set.

Exercise A 1.126. Find a set M such that $M = Dr(M)$.

Exercise A 1.127. Find a set M such that $M = Dr(M)$ and M is totally disconnected.

Exercise A 1.128. Find a set M such that $M = Dr(M)$ and M is compact and totally disconnected.

Definition A 1.129. If M is a point set in the space S , then S is said to be **limit compact** iff every infinite subset of M has a limit point in M .

Theorem A 1.130. No non-degenerate locally compact, closed, and connected set is the union of countably many mutually exclusive compact point sets.

Theorem A 1.131. If M is compact, then M is limit compact.

Theorem A 1.132. If M is closed and limit compact, then M is compact.

Exercise A 1.133. Find a T_3 - space that does not satisfy theorem A 1.132.

Exercise A 1.134. Find a T_2 - space that does not satisfy theorem A1.132.

Theorem A 1.135. No non-degenerate locally compact closed and connected point-set is the union of countably many mutually exclusive compact point-sets.

Theorem A 1.135 Restated: No non-degenerate locally compact closed and connected point-set is the union of a countable non-degenerate collection of mutually exclusive compact point-sets.

Chapter 2

A Tad More Concepts in Point-Set Topology

2.1 Product Spaces

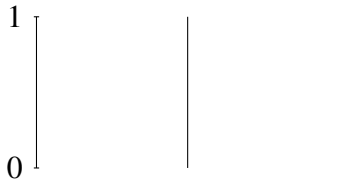
Definition A 2.1. Suppose that I is a set and for each $i \in I$ X_i is a topological space. The **product space** of the spaces $\{X_i\}_{i \in I}$ is the set to which x belongs iff there exists a function from I into $\bigcup_{i \in I} X_i$ so that $x(i) = x_i (x_i \in X_i)$ for each $i \in I$.

Example A 2.2. The idea of $X_1 \times X_2$. $I = \{1, 2\}$. $S = X_1 \times X_2$. $x \in S$. x is (x_1, x_2) where $x_1 \in X_1$ and $x_2 \in X_2$.
 $x : I \longrightarrow X_1 \cup X_2$.

Notation A 2.3. Let S be the product space of I sets X_i where $i \in I$. The notation is $S = \prod_{i \in I} X_i$.

Definition A 2.4. R is a region for S means that there is a finite subset $I' = \{i_1, i_2, \dots, i_n\}, n \in \mathbb{N}$ if I and for each $i_\ell \in I'$ there is a region R_{i_ℓ} of X_{i_ℓ} so that $R = \{x \in S | x_{i_\ell} \in R_{i_\ell} \text{ for all } i_\ell \in I'\}$.

Example A 2.5. Let $U = \mathbb{R} \times \mathbb{R}$. $X_i = [0, 1]$ for $i = 1, 2, 3$. $Y = [0, 1]$.
 $X_i \times Y$ for $i = 1, 2, 3$ $((X_1 \times Y) \cup (X_2 \times Y) \cup (X_3 \times Y))$.



Definition A 2.6. (The Box Topology: $S^\square$). R is said to be a region for $S^\square$ means that for each $i \in I$ there is a region R_i in X_i so that $R = \{x | x_i \in R_i\}$.

Theorem C 2.7. $\exists$ a well-ordering on $\mathbb{R}$ so that every initial segment (with respect to that ordering) is uncountable.

Axiom CH. Let S be a space. If M is infinite and less numerous than $\mathbb{R}$ then M is denumerable.

Axiom $\neg\text{CH}$. Let S be a space. There exists an uncountable set such that it is less numerous than the reals.¹

2.2 A Bit More About Continua

Definition D 2.8. If M is a continuum, then

1. p is said to be a **cut point** of M means that $M \setminus \{p\}$ is not connected; whereas,
2. q is said to be a **non-cut point** of M means that $M \setminus \{q\}$ is connected.

Definition D 2.9. If M is a continuum and p is a point of M , then **the composant of M at p** is the set to which x belongs iff there is a proper sub-continuum of M containing x and p .

Notation D 2.10. Let the composant of M at p be denoted as $C_p(M)$.

Definition D 2.11. The point-set M is **nowhere dense** in the space S means that if $p \in S$ and U is an open set containing p , then there is an open set V which is a subset of U where $V \cap M = \emptyset$.²

Definition D 2.12. If M is a continuum and N a point set, then M is called a **continuum of condensation of N** if M is a non-degenerate subset of N and every point of M is a limit point of $N \setminus M$.

Definition D 2.13. If M is a continuum, then M is called an **indecomposable continuum** if M is non-degenerate but is not the union of two continua both distinct from it.

¹This is stating the 'not continuum hypothesis' as an axiom (in other words one oft either assumes the CH or the $\neg CH$). Many claims do not need either and are 'free' of the CH vs. $\neg CH$ question. However, this demonstrates the set-theoretic disparity of advanced set theory, topology, analysis, etc. studies.

²Meaning, of course, M has no interior. Notice the point p is arbitrary - it is the fact that a point set M is said to be nowhere dense if and only if each region contains a region that does not intersects M .